

From Cash to Buy-Now-Pay-Later: How Platform Credit Shapes Market Allocation^{*}

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September 2, 2026

Abstract

Online marketplaces increasingly supply the buy-now-pay-later credit used at their own checkouts. We show that a marketplace levying the same ad valorem fee on both payment methods over-supplies credit, even when it provides credit at cost. The reason is a difference in demand elasticities. A pay-now buyer must satisfy a liquidity constraint that a pay-later buyer escapes, so pay-now demand is more elastic. An ad valorem fee costs a seller a share of profit equal to her demand elasticity, so a common rate taxes pay-now more heavily and implicitly subsidizes pay-later. Sellers therefore adopt credit beyond the laissez-faire margin, at every fee level, for every liquidity distribution, and even with fully attentive buyers. Naive buyers lower pay-later elasticity further, amplifying the platform's tilt, and they accept deferral they would refuse if they saw its full cost. Disclosure addresses misperception; only the fee structure across payment methods addresses the platform's distortion.

Keywords: *Platforms, Buy-Now-Pay-Later, Liquidity Constraints, Naivety*

^{*}This article has benefited from the comments by Fabrizio Mattesini, Shengxing Zhang, Yu Zhu, Timothy Kam, and participants in Kyoto Workshop on Digitalization and Macroeconomic Policies (2024), Singapore Search and Matching Workshop (2024), Summer Workshop on Money, Banking, Payments, and Finance (2024). The remaining errors are ours. Hu acknowledges the financial support from the Natural Science Foundation of China (Grant 72003041), the Shanghai Pujiang Program (Grant 21PJJC011), and the Shanghai Institute of International Finance and Economics. Watanabe acknowledges the financial support from JSPS KAKENHI (Grants JP23H00054, JP22K20161, and JP23K17286) and the Murata Science Foundation.

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1 Introduction

E-commerce platforms have evolved beyond matching buyers and sellers; they now also set the payment services available at checkout. A prominent service they offer is “Buy Now, Pay Later” (BNPL), which lets consumers take the good immediately and defer payment for several weeks or even months. Platforms supply this option themselves or through a partner they bring to their own checkout: Amazon and Walmart offer pay-over-time globally, Otto and MediaMarktSaturn offer purchase-on-invoice and installment plans in Germany, and JD.com and Alibaba’s marketplaces extend deferral through Baitiao and Huabei in China.

Regulatory frameworks have so far treated BNPL as a consumer-credit problem, and their focus has been consumer education. The United Kingdom brings BNPL inside the Financial Conduct Authority’s perimeter in July 2026, mandating disclosure of repayment terms and affordability checks; the European Union’s revised Consumer Credit Directive, in force in November 2026, standardizes cost disclosure; and Australia has regulated BNPL as credit under the National Credit Act since June 2025. What these regimes share is that they all work on what the buyer sees and can carry. None of them asks what changes when the marketplace itself is the one supplying the credit.

This paper asks that question. A platform that supplies pay-later credit under the fee schedules that practice and regulation deliver puts more of it into the market than *laissez-faire* would, and it does so even when every buyer understands exactly what deferral will cost her — the late fees, the penalty charges, the mark left on her credit record.

The force behind the result is a gap in demand elasticities that the liquidity constraint opens between the two payment methods. Call them *money* and *credit*: paying with money means settling on the spot, paying with credit means settling later. A buyer at a money seller must clear two hurdles, then — she must want the good, and she must have the funds on hand when she checks out. Raise the price and the seller loses buyers at both: some no longer want the good, and some who still want it can no longer settle. Credit removes the second hurdle. Money demand is therefore more elastic than credit demand, and the liquidity constraint is itself the source of that elasticity.

Now let the platform supply credit at marginal cost and charge the same net *ad valorem* fee on both payment rails, so that nothing in the schedule reads as a subsidy to credit. Platforms in practice usually price credit below cost to win merchant adoption, and whatever over-supply an equal fee produces, a subsidized fee produces more of. Even here, more sellers accept credit than

would accept it in the laissez-faire economy. The reason is that a fee charged on revenue is paid out of margin, and the two differ in size by exactly the demand elasticity. A seller pricing optimally leaves a margin equal to the fraction $1/\varepsilon$ of her revenue, where ε is the elasticity she faces, so one percentage point of revenue costs her ε percent of profit: the more elastic her demand, the thinner the margin sitting under each dollar of sales, and the more of her profit a given headline rate takes. Money is the more elastic method, so an equal rate bites harder there, and sellers near the adoption margin tilt onto credit. The result needs nothing about the buyer's psychology: the platform over-supplies credit even when buyers are perfectly attentive.

None of this denies that consumer naivety matters. It matters, and it presses on the very same lever. A naive buyer responds less to the posted price when she pays later: the more naive the buyer, the less elastic credit demand, the wider the gap between the elasticities of the two payment rails, and the more the platform's fee tilts sellers onto credit.

The rest of the paper makes this argument precise, and it does so in three steps. We first build an economy with no platform in it, to establish which sellers adopt credit when nothing distorts the choice. We then introduce the platform and its fees, which is where the tilt appears. Finally we let buyers misperceive what deferral costs them, which is where the second distortion appears.

Section 2 builds the microfoundation. A continuum of sellers each supply a unique good, vertically differentiated by quality, and set its price; each faces her own pool of buyers who want at most one unit. Buyers differ along two dimensions: how much they value the good, and how much cash they have available for the purchase. Every seller accepts money, and a money sale needs two things to go right — the buyer must want the good at the posted price, and she must be able to settle on the spot. Demand is therefore the fraction of buyers who want the good times the fraction who can pay for it.

A seller can choose to accept credit. Credit is costly, in that the technology incurs a fraction of the transaction, and what the seller buys with that cost is the removal of the liquidity constraint. Adoption thus comes down to a single tradeoff — pay a real resource cost to recover the buyers the liquidity constraint was excluding — and whether the trade is worth making depends on how many buyers that constraint excludes. This is where quality enters. A higher-quality good commands a higher willingness to pay and so carries a higher dollar price, and a higher dollar price is one that fewer buyers can settle from cash on hand. Nothing turns on quality as such: any attribute that raises both what buyers will pay and what the good costs to produce — technical

performance, durability, brand, bundle size — does the same work, since all that matters is that the transaction is larger in dollars. Money profit therefore falls with quality while credit profit does not, and adoption sorts cleanly by quality: above a threshold sellers accept credit, below it they accept money only. We characterize that threshold and compare it with the choice of a constrained planner, who decides which sellers accept credit but takes sellers' pricing as given.

Building on this microfoundation, in Section 3 we construct an economy in which a monopolistic platform operates the market. The platform's matching technology brings each seller a larger pool of buyers than the outside market does, and this advantage is what allows it to charge positive fees. It levies *ad valorem* fees on the two payment options: a net fee s_M on money and a net fee s_B on credit. We start with the equal net fee, $s_B = s_M$, and credit provided at cost. As illustrated above, money demand is the more elastic, so an equal net fee taxes the money rail more than credit and tilts sellers onto credit. The result holds at every fee level, for every distribution of liquidity, and even when buyers are fully attentive. When willingness to pay has constant elasticity, the allocation solves in closed form: the fee moves the adoption margin down by exactly that rate.

We then let the two fees differ. What decides the direction of the allocation is not either fee on its own but the pair: a fee destroys a share of the marginal seller's profit equal to her demand elasticity times the fee, so the fee pair over-supplies credit exactly when the credit fee falls below a threshold curve in the fee plane — the curve at which the fee's bite on the two rails is equal. This curve, the allocation-neutral frontier, bounds a whole region of fee pairs that over-supply credit; the equal-fee diagonal lies strictly inside it, which is the equal-net-fee result as a special case. Over-supply is thus not a knife-edge at equal fees but a region of the fee plane.

We finally ask where the platform's own profit-maximizing fees land. To a first approximation the platform prices each rail by the inverse-elasticity rule: it taxes each rail at the rate inverse to how elastically that rail's seller revenue responds to the fee, so the less elastic base pays the higher rate. Credit is the less elastic base — its demand carries no liquidity elasticity — so the rule taxes credit at the higher net rate. The revenue logic, however, runs against the allocation logic: a fee's bite on a seller's profit is proportional to the demand elasticity she faces, so the money rail, being the more elastic, is still taxed harder per dollar of profit even at the lower rate. The two forces net out in favor of over-supply when the marginal adopter's liquidity elasticity exceeds the money pool's average — that is, when liquidity elasticity rises with the size of the transaction; when it is constant across the pool, so that the marginal adopter is representative of the pool, the direction reverses and the platform under-supplies. Nor does the exact optimum escape the region: every

fee pair on one side of the allocation-neutral frontier gains from reassigning the marginal seller between the rails, and that gain pushes the fees toward the frontier but never across it, so the direction the inverse-elasticity rule sets is the direction at the optimum.

In Section 4, we relax the assumption that buyers correctly perceive the cost of deferral. A naive buyer makes two mistakes. First, she underestimates the hidden load, so credit looks cheaper than it is: a level error. Second, she fills the missing part with a stored impression that does not move with the posted price, so her demand for credit becomes less elastic: an elasticity error. The level error drives a wedge between the market and the planner: buyers who would regret the trade take credit anyway, and the market's adoption margin can move beyond the planner's. The elasticity error is what our framework contributes, and it lands on the platform's own instrument: a fee's bite is proportional to the demand elasticity, so less elastic credit demand further widens the elasticity gap between the two payment methods and steepens the platform's tilt the more naive buyers are. Naivety, therefore, works the platform's own lever, and the over-supply of credit compounds.

Section 5 concludes by mapping these two margins onto the two instruments and onto the regulatory regimes now coming into force. We show that the informational/naivety margin can be closed by disclosure that is written per transaction and denominated in the price; a generic warning moves the buyer's stored impression rather than its responsiveness to the price. The platform margin (fee structure) survives even with fully attentive buyers; a cap on the level shrinks the misallocation — the over-supplied band contracts one-for-one with the fee — but only the tilt between the two options closes it. In particular, a rule capping only positive credit fees never reaches the subsidy corner, where platforms price credit below cost to drive merchant adoption. The regulation now arriving is a well-aimed attack on the informational margin; it leaves the platform margin untouched.

Related Literature

Platform fees and payment cards. The paper sits closest to work on payment-card fees and two-sided platforms, which asks how an intermediary splits its fee between merchants and consumers and whether the privately chosen split is socially right (Rochet and Tirole, 2002, 2003; Armstrong, 2006; Wright, 2004; Bedre-Defolie and Calvano, 2013; Edelman and Wright, 2015). We model the fee as *ad valorem* because marketplaces post *ad valorem* take rates and card networks charge proportional fees, a practice Shy and Wang (2011) rationalize; Wang and Wright

(2017) show that a linear ad-valorem fee earns the platform what it would earn by setting the optimal fee for each good, as if it observed the goods' unobserved costs and values.

Ad valorem taxation and incidence. That an ad valorem fee erodes profit at the rate of the demand elasticity is standard incidence theory under imperfect competition (Suits and Musgrave, 1953; Delipalla and Keen, 1992; Anderson et al., 2001; Weyl and Fabinger, 2013). We take the result and put it to a different use. The object being taxed here is a binary technology choice rather than a quantity, and the two elasticities differ because a liquidity constraint binds on one rail and not on the other — so incidence determines an allocation, not merely a division of surplus.

Inattention, shrouded attributes, and naive borrowing. The elasticity gap has a second source. Chetty et al. (2009) established that a cost component consumers process incompletely moves behavior less than one for one; Finkelstein (2009) and Taubinsky and Rees-Jones (2018) find the same pattern in other settings, and Gabaix (2014) supplies the decision-theoretic form we adopt, with Bordalo et al. (2013) a close relative. Gabaix and Laibson (2006) and Heidhues and Köszegi (2010, 2017) study the level counterpart, in which buyers underestimate hidden add-ons and future repayment costs and firms price against that error. This literature asks what misperception does to what buyers purchase and what they pay for it. It does not ask what misperception does to the seller's choice of payment technology, which is our question, and the answer runs through the same object the platform's fee operates on. A buyer whose perceived cost does not track the posted price has an inelastic demand for the rail that carries the cost, so the fee spares that rail. Misperception and the liquidity screen turn out to be two sources of one asymmetry.

Buy-now-pay-later. The empirical work on the product supplies our motivation and our closest challenge. Di Maggio et al. (2022) find that access raises spending and, among liquidity-constrained users, overdraft fees, which is evidence that the deferral relieves a real constraint and carries a real cost. Bian et al. (2023) document the platform-embedded form of the product, the case we model. Berg et al. (2025) come closest to our adoption margin: pay-later raises merchant sales by 20 percent, concentrated among low-creditworthiness buyers and in products where market power is larger, and the merchant's benefits exceed the costs. They read the product as a price-discrimination device, and on their question — the private return to adoption — we have no quarrel. Our comparison is between that private return and a constrained planner's, and the two diverge for a reason a merchant-side design cannot see: the platform's fee prices the

rails by their elasticities rather than by their social costs. A merchant can be right to adopt and the allocation still be wrong. Survey and regulatory evidence on what buyers know about the cost of deferral is summarized in Financial Conduct Authority (2021) and Consumer Financial Protection Bureau (2022).

2 The Environment and the Laissez-Faire Economy

2.1 Agents

There is a unit mass of sellers indexed by product quality ξ , distributed as G with density g on $[\xi_\ell, \xi_h]$. Each seller supplies a unique good and is therefore a monopolist. A seller with indexed by ξ has a constant marginal cost $c\xi$.

Each seller meets a pool of potential buyers with measure normalized to one. Each buyer is indexed by a pair (θ, y) : θ is her willingness to pay per unit of quality, drawn from F on $[0, \theta_h]$ with $\theta_h \leq \infty$, and y is the cash she has available for the purchase this good, drawn from H with a support of $[0, \infty)$. The two components are independent of each other and of ξ .¹

Each buyer demands at most one unit, and a buyer who acquires a quality- ξ good at the dollar price p obtains utility of

$$\theta\xi - p.$$

The multiplicative valuation of $\theta\xi$ follows from the literature of vertical-differentiation (Mussa and Rosen, 1978; Gabszewicz and Thisse, 1979; Shaked and Sutton, 1982, 1983).

We take the quality of sellers as given. In fact, the interpretation of ξ can be more general than quality. ξ scales both willingness to pay and cost, so a higher ξ simply identifies a more valuable, more expensive good. Any attribute with that property — technical performance, durability, brand, bundle size — fits the model. What matters is the consequence: a higher ξ means a larger dollar transaction, and more buyers cannot settle that from cash on hand.

2.2 Payment and Demand

Every seller can accept pay-now, or money; accepting credit — buy now, pay later — is an option she chooses. Money is shorthand for any payment method that requires the buyer to have

¹Because each seller meets her own pool of buyers and each buyer in that pool is deciding only whether to buy that seller's good, y is the cash the buyer has available for this particular purchase, not a budget to be allocated across multiple sellers or purchases.

the full transaction amount available in her account at the moment of purchase: currency, debit, or any prefunded balance. Credit is shorthand for any pay-later method: the buyer takes the good and settles some other day, so no cash is needed at the point of sale.

Credit differs from money in the following ways. It removes the cash constraint of buyers at the point of sale. It costs real resources: the credit technology absorbs a fraction ω of the transaction amount. And it carries a *hidden load* $\mu > 0$ per dollar deferred — late fees, penalty charges, credit-record damage — against a *deferral benefit* $\beta \in (0, 1)$ per dollar deferred, which captures the free loan over the installment window, the buffer left intact, the shorter checkout. We assume $\beta > \mu$, so that deferral is worth having once its cost is correctly understood.

Perception. The deferral benefit is visible at checkout. The load is not fully so. A buyer perceives a fraction $v \in [0, 1]$ of it and fills the remainder with a stored, fixed impression k of what pay-later costs on a purchase like this one, so against a normalized price z she evaluates credit at the perceived all-in price

$$\tilde{z} = (1 - \beta + v\mu)z + (1 - v)k. \quad (2.1)$$

Because $\beta > \mu$ gives $\tilde{z} < z$, every buyer at a credit-accepting seller prefers to defer, whether or not she has the liquidity to pay on the spot. Surcharging credit is not allowed, so she pays the same p either way, and a seller who accepts credit therefore sells to all her buyers on that rail.

This section and the next section impose $v = 1$, so that buyers perceive the load correctly, the stored impression carries no weight, and $\tilde{z} = (1 - \beta + \mu)z$. We impose in addition $\beta - \mu \rightarrow 0^+$, so that in the equations $\tilde{z} \rightarrow z$: a buyer is then indifferent between the rails on price alone, and takes credit for the liquidity it frees. Section 4 relaxes $v = 1$ and lets the load and the belief do their work.

The two payment methods generate different demands from the same primitives. Throughout, let p be the dollar price, and $z \equiv p/\zeta$ the quality-normalized price. Under **credit payment**, only the want matters: the buyer settles later, so the cash constraint disappears and the sole hurdle is whether she wants the good at the price she perceives, $\theta \geq \tilde{z}$. Under the benchmark restriction $\tilde{z} = z$, so demand is independent of the liquidity distribution:

$$d_B(z) = \bar{F}(z).$$

Under **money payment**, the buyer must in addition be able to pay for the good on the spot, $y \geq p$,

so demand is

$$d_M(z, \xi) = \bar{F}(z) \bar{H}(p).$$

The factor $\bar{H}(p)$ is the **liquidity factor**: the fraction of buyers who can pay the dollar price. It falls as the price rises, so it adds to the demand elasticity: it excludes more buyers from the money rail as p increases.²

Throughout the paper we use pay-later and credit interchangeably, and note that a credit card relaxes the same liquidity constraint, so the reader may equally read the credit rail as a card rail without changing the mechanics. We frame the paper around BNPL because the two objects we study are sharper there. First, acceptance is a live margin: in e-commerce-heavy economies, developed and developing alike, whether a seller takes a pay-later method is typically an interior decision — some sellers do, some do not — while cards are already universal. Second, the platform controls the credit fee: BNPL fees are set by the marketplace, whereas card merchant discounts are largely determined by interchange and outside the platform’s control. BNPL is also a focal point of regulatory concern: it is used almost exclusively online, and its users are disproportionately young — the buyers most prone to impulse purchases and to costly default.

In the following, we compare three allocations built on the same primitives: the *laissez-faire economy*, the *planner’s allocation*, and the *platform economy*.

2.3 The Laissez-Faire Economy

In the laissez-faire economy, money is free and credit is supplied competitively at its resource cost, a fraction ω of the transaction amount. Each seller makes one binary choice: accept credit or take money only. She accepts credit when it is more profitable. Which sellers are those? The higher-quality ones, because they charge higher prices and therefore have more to gain from relaxing their buyers’ cash constraint. We first characterize the prices on the two rails.

Seller Pricing. It is convenient to solve the seller’s pricing problem once, for a general ad valorem fee x levied on transaction value. Laissez-faire is the case $x = 0$ on money and $x = \omega$ on credit. A fee at rate x leaves the seller $1 - x$ of each revenue dollar against a cost of $c\xi$, so it acts as a tax on cost: it turns unit cost c into effective cost $a \equiv c/(1 - x)$.

Under credit, no buyer is cash constrained: demand is $\bar{F}(z)$, and the seller’s profit-maximizing

²Both rails are viable: because H has unbounded support, $\bar{H}(p) > 0$ at every dollar price and money payment is feasible for every seller, however high the price; and because $c/(1 - \omega) < \theta_h$, credit can be offered at a positive margin with positive demand. Neither assumption binds beyond guaranteeing the interior allocation we study; the second is vacuous when $\theta_h = \infty$.

problem is

$$\max_z [(1-x)z - c] \bar{F}(z) = (1-x) \max_z (z - a) \bar{F}(z).$$

The factor $1 - x$ is a constant and does not affect the price: she prices exactly like a monopolist with cost a . We divide the objective by ξ , so all profits below are per unit of quality; total profit is the per-unit amount times ξ . The first-order condition is:

$$\frac{z^*(a) - a}{z^*(a)} = \frac{1}{\lambda(z^*(a))}.$$

Here $\lambda(z) \equiv \frac{zf(z)}{\bar{F}(z)} = -d \ln \bar{F}(z) / d \ln z$ is the elasticity of credit demand and the generalized failure rate of F : a 1% price rise makes λ percent of buyers walk away. Let $z_B = z^*(a)$ and define revenue per unit of quality by $R_B(z) \equiv z \bar{F}(z)$. Then *pre-fee profit per unit of quality* is

$$\pi_B(a) = (z_B - a) \bar{F}(z_B) = \frac{R_B(z_B)}{\lambda(z_B)}.$$

So the seller keeps $(1-x)\pi_B(a)$.

Under money, the liquidity factor enters and the seller solves

$$\max_z (z - a) \bar{F}(z) \bar{H}(z\xi),$$

up to the same constant $1 - x$. Demand is now more elastic: a price increase loses buyers both because some no longer want the good and because some can no longer pay for it. The Lerner condition becomes:

$$\frac{z_M - a}{z_M} = \frac{1}{\lambda(z_M) + \gamma(p_M)},$$

where $p_M \equiv z_M \xi$ is the dollar price and $\gamma(p) \equiv ph(p) / \bar{H}(p)$ is the generalized failure rate of H : a 1% rise in the dollar price shuts γ percent of currently solvent buyers out of their wallets. The money seller's elasticity is therefore $\lambda + \gamma$, against the credit seller's λ .

Profit per unit of quality before the fee is

$$\pi_M(a, \xi) = (z_M - a) \bar{F}(z_M) \bar{H}(p_M) = \frac{R_M(z_M, \xi)}{\lambda(z_M) + \gamma(p_M)},$$

where $R_M(z_M, \xi) \equiv z_M \bar{F}(z_M) \bar{H}(p_M)$ is money revenue per unit of quality.

A higher elasticity means a thinner mark-up, so at a common effective cost $z_M \leq z^*(c)$. Under laissez-faire the credit seller also faces the higher effective cost $c/(1-\omega) > c$, so $z_B = z^*(c/(1-\omega)) > z^*(c)$. Together,

$$z_M \leq z^*(c) \leq z_B.$$

Who Adopts Credit? An increase in ξ raises the dollar price $p_M = z_M \xi$, so more buyers cannot meet the cash need, and the money rail becomes less profitable. The elasticity $\gamma(p_M)$ is therefore the profit erosion rate: the more binding the liquidity constraint at the seller's price, the faster money profit falls as quality rises.

Lemma 1 *For every effective cost a ,*

$$\frac{\partial \ln \pi_M(a, \xi)}{\partial \ln \xi} = -\gamma(p_M) < 0.$$

Proof. By the envelope theorem $\partial \pi_M / \partial \xi = -(z_M - a) \bar{F}(z_M) h(p_M) z_M$. Multiplying by ξ / π_M leaves $-p_M h(p_M) / \bar{H}(p_M)$. ■

Credit profit is flat in ξ while money profit strictly falls in it, so the two cross at most once; that crossing is the market's adoption margin.

Proposition 1 *Suppose*

$$\pi_M(c, \xi_h) < (1 - \omega) \pi_B\left(\frac{c}{1 - \omega}\right) < \pi_M(c, \xi_\ell).$$

Then there is a unique $\xi^ \in (\xi_\ell, \xi_h)$ solving*

$$\underbrace{(1 - \omega) \pi_B\left(\frac{c}{1 - \omega}\right)}_{\text{credit profit, flat in } \xi} = \underbrace{\pi_M(c, \xi^*)}_{\text{money profit, falling in } \xi}, \quad (2.2)$$

and every seller with $\xi > \xi^$ accepts credit while every seller with $\xi < \xi^*$ takes money only.*

A worked *Pareto-Pareto* case carries the logic further. Suppose both F and H are Pareto. Let

$$\bar{F}(z) = \min\{1, (z/\theta_0)^{-\varepsilon}\}, \quad \bar{H}(p) = \min\{1, (p/y_0)^{-\alpha}\},$$

with $\varepsilon > 1$ and $\alpha > 0$.³ Both failure rates are now constant, $\lambda \equiv \varepsilon$ and $\gamma \equiv \alpha$, so the Lerner conditions solve in closed form, $z_B = \frac{c}{1-\omega} \cdot \frac{\varepsilon}{\varepsilon-1}$, $z_M = c \cdot \frac{\varepsilon+\alpha}{\varepsilon+\alpha-1}$, and quality enters pricing only through the dollar price $p_M = z_M \xi$. Equating the retained profits in (2.2) pins down the liquidity factor of the marginal seller,

$$\bar{H}(z_M \xi^*) = K \equiv \frac{\varepsilon + \alpha - 1}{\varepsilon - 1} \left(\frac{z_M}{z_B}\right)^\varepsilon, \quad \xi^* = \frac{y_0}{z_M} K^{-1/\alpha}.$$

³Here θ_0 and y_0 are the lower supports: below θ_0 every buyer values the good above the price, and below y_0 every buyer can pay; above those cutoffs, the Pareto tails apply. We set $\theta_h = \infty$. Because every equilibrium price exceeds effective cost $a \geq c$, the condition $c \geq \theta_0$ keeps all relevant prices in the constant-elasticity region; similarly, we restrict attention to prices above y_0 . $\varepsilon > 1$ keeps surplus integrals finite.

The adoption margin is interior exactly when $\left(\frac{z_M \xi_h}{y_0}\right)^{-\alpha} < K < \left(\frac{z_M \xi_\ell}{y_0}\right)^{-\alpha}$. Since K is a survival probability, $K > 1$ means no threshold exists: no seller stays on money. Free credit is a special case: at $\omega = 0$, credit removes the liquidity constraint at no cost and strictly dominates money for every seller, so there is no interior margin.

2.4 Laissez-Faire against the Planner

We now compare the laissez-faire allocation with a constrained social planner. The planner chooses which sellers accept credit, but she takes the sellers' monopoly pricing as given. She can reassign sellers across payment rails; she cannot reset prices.

Consider a seller of quality ξ . If assigned to credit, she prices at z_B , every buyer who wants the good trades, and each trade costs society the rail's resource cost ωz_B plus the load μz_B the buyer bears, less the deferral benefit βz_B she enjoys. The net social cost of a credit dollar is therefore $\kappa \equiv \omega + \mu - \beta$, which the benchmark restriction $\beta - \mu \rightarrow 0^+$ reduces to ω , so that per unit of quality

$$W_B = \int_{z_B}^{\theta_h} (\theta - c - \omega z_B) dF(\theta).$$

If assigned to money, she prices at z_M , a buyer trades only if she wants the good and can pay for it, and surplus per unit of quality is

$$W_M(\xi) = \bar{H}(p_M) \int_{z_M}^{\theta_h} (\theta - c) dF(\theta).$$

W_B is flat in ξ , exactly as credit profit was, so single crossing requires only that W_M fall. Two opposing effects operate on W_M . A higher ξ raises the dollar price and shuts buyers out, lowering surplus; but it also induces the seller to cut her normalized price, which raises the surplus of those who remain. The condition below ensures the first effect dominates.

Lemma 2 *Let $\psi(\xi) \equiv -d \ln z_M / d \ln \xi$ be the rate at which the money seller cuts her normalized price as quality rises. If $\psi < \gamma(p_M) / [\gamma(p_M) + \lambda(z_M)]$, then W_M is strictly decreasing, and the planner's optimal allocation is a threshold: she assigns money to sellers with $\xi < \xi^{**}$ and credit to those with $\xi > \xi^{**}$, where ξ^{**} solves $W_M(\xi^{**}) = W_B$.*

Proof. Since $d \ln p_M / d \ln \xi = 1 - \psi$, differentiating $W_M = \bar{H}(p_M) \int_{z_M}^{\theta_h} (\theta - c) dF(\theta)$ gives

$$\frac{d \ln W_M}{d \ln \xi} = -\gamma(p_M) (1 - \psi) + \frac{z_M(z_M - c)f(z_M)}{\int_{z_M}^{\theta_h} (\theta - c) dF(\theta)} \psi.$$

The first term is the liquidity effect, negative whenever $\psi < 1$; the second is the price-restraint

effect, with the sign of ψ . Its coefficient is at most $\lambda(z_M)$, because $\int_{z_M}^{\theta_h} (\theta - c) dF(\theta) \geq (z_M - c)\bar{F}(z_M)$. Thus for $\psi \leq 0$ both terms are nonpositive and the first is strictly negative; for $\psi > 0$ the derivative is at most $-\gamma(1 - \psi) + \lambda\psi$, negative exactly under the condition. ■

The condition $\psi < \gamma(p_M)/[\gamma(p_M) + \lambda(z_M)]$ asks that the money price not respond too elastically to quality.⁴ In the Pareto–Pareto setup of Section 2.3, $\lambda \equiv \varepsilon$ and $\gamma \equiv \alpha$, $z_M = c \frac{\varepsilon + \alpha}{\varepsilon + \alpha - 1}$, which does not depend on ξ , so $\psi = 0$. The condition then reduces to $0 < \frac{\alpha}{\alpha + \varepsilon}$, which always holds.

Having characterized the planner’s allocation, we now compare it with the market’s. Write $z_M \equiv z_M(\xi^*, c)$ for the marginal seller’s money price, $p_M^* \equiv z_M \xi^*$ for the dollar price it implies, and $S(z) \equiv \int_z^{\theta_h} (\theta - z) dF(\theta)$ for buyer surplus at price z .

Proposition 2 *Under the condition of Lemma 2, laissez-faire over-supplies credit — that is, $\xi^* < \xi^{**}$ — if and only if*

$$\bar{H}(p_M^*) S(z_M) > S(z_B). \quad (2.3)$$

Proof. Split each surplus into buyer surplus and seller profit: $W_M(\xi^*) = \bar{H}(p_M^*) S(z_M) + (z_M - c)\bar{F}(z_M)\bar{H}(p_M^*)$ and $W_B = S(z_B) + [(1 - \omega)z_B - c]\bar{F}(z_B)$. The indifference condition (2.2) equates the two profit terms, so $W_M(\xi^*) - W_B = \bar{H}(p_M^*) S(z_M) - S(z_B)$. By Lemma 2, $W_M - W_B$ is strictly decreasing and vanishes at ξ^{**} , so its sign at ξ^* is the sign of $\xi^{**} - \xi^*$. ■

The criterion is a race between two effects. Credit relieves the liquidity constraint: it serves every buyer who wants the good, whereas money reaches only the fraction $\bar{H}(p_M^*) < 1$ who can pay on the spot. But credit also strengthens the seller’s market power. By the tie-breaking assumption, a credit-accepting seller expects her buyers to pay on credit, so she faces the less elastic credit demand and charges $z_B > z_M$, and the higher price is what each buyer loses. Laissez-faire over-supplies credit exactly when consumers suffer from the marginal seller’s adopting credit rather than money, which (2.3) rewrites as

$$\underbrace{S(z_M) - S(z_B)}_{\text{market-power effect}} > \underbrace{(1 - \bar{H}(p_M^*)) S(z_M)}_{\text{liquidity-relief effect}}.$$

Reversed, the market under-supplies credit. The left side is the surplus the higher credit price destroys; the right side is the surplus credit creates by serving the buyers money excludes. The seller, indifferent between the two rails at ξ^* , ignores both; the planner counts both. When the two effects are equal, market and planner face the same marginal incentives.

⁴The condition is sufficient, not necessary. The exact coefficient on the price-restraint effect is $A \equiv z_M(z_M - c)f(z_M)/\int_{z_M}^{\theta_h} (\theta - c) dF(\theta) \leq \lambda(z_M)$, so the exact condition for W_M to be decreasing is $\psi < \gamma(p_M)/[\gamma(p_M) + A]$. Since $A \leq \lambda(z_M)$, the lemma’s condition implies the exact one, but the converse need not hold.

Under the Pareto–Pareto assumption the criterion takes a closed form: with $\lambda \equiv \varepsilon$, buyer surplus is $S(z) = z\bar{F}(z)/(\varepsilon - 1)$, so it reads $\bar{H}(p_M^*) > z_B\bar{F}(z_B) / z_M\bar{F}(z_M)$. The demand ratio on the right is not free: indifference pins it down. Inserting the two retained mark-ups $z_M - c = c/(\varepsilon + \alpha - 1)$ and $(1 - \omega)z_B - c = c/(\varepsilon - 1)$ into (2.2) gives

$$\frac{\bar{F}(z_B)}{\bar{F}(z_M)} = \frac{\varepsilon - 1}{\varepsilon + \alpha - 1} \bar{H}(p_M^*),$$

and substituting this into the criterion cancels the liquidity factor from both sides. Inserting the two prices yields

$$\omega < \frac{\alpha}{\alpha + \varepsilon}. \quad (2.4)$$

This condition says credit is cheap relative to the strength of the liquidity elasticity — ω below $\alpha/(\alpha + \varepsilon)$, a threshold that rises with α . Why does a strong liquidity elasticity produce too much credit? A large α makes the marginal money seller’s demand highly elastic: liquidity adds α to her elasticity, and by the Lerner condition a higher elasticity means a lower price z_M and hence more buyer surplus $S(z_M)$. The seller does not count that surplus — indifference holds her profit equal to credit profit, which is independent of α , so the low price costs her nothing. The surplus created by liquidity is a pure gift to money buyers, and the market ignores gifts when choosing rails; the stronger the liquidity elasticity, the more buyer surplus money generates per dollar of seller profit, and the more the market under-supplies money.

3 The Platform Economy

3.1 The Monopoly Platform

A platform now sits between buyers and sellers. Its only departures from the laissez-faire economy are a matching advantage and a fee pair. The platform’s matching technology brings each seller a pool of measure $m > 1$ buyers, against the unit measure the seller reaches in the outside market; this advantage is what sustains positive fees through seller participation. Sellers choose between the platform and the outside market — the laissez-faire economy of the previous subsection — and buyers move with their sellers. Note that the matching advantage does not move the rail-choice margin, because m multiplies profits on both rails equally and cancels from the money-versus-credit comparison.

The platform charges *ad valorem* fees: s_M on money, and a gross rate $\tilde{s}_B$ on credit trade, where

it bears the credit cost ω itself and nets $\tilde{s}_B - \omega$. We work with the net credit fee

$$s_B \equiv \frac{\tilde{s}_B - \omega}{1 - \omega} \approx \tilde{s}_B - \omega,$$

so $s_M = s_B$ means equal net margins on the two rails. The fee pair (s_M, s_B) lies in $\mathcal{S} \equiv [0, \bar{s}_M(m)] \times [0, \bar{s}_B(m)]$ where the upper bounds are positive functions of m .

We can read the game in two stages: the platform first announces (s_M, s_B) , and each seller then decides whether to join after observing the fees. A seller who stays out earns the laissez-faire profit, the larger of $\pi_M(c, \xi)$ and $(1 - \omega)\pi_B(c/(1 - \omega))$; a seller who joins earns her best profit after paying fees to the platform (see below). The participation constraint — that joining weakly dominates staying out — pins down the upper bounds of $\mathcal{S}$.

We consider various fee schedules the platform may post; for most of our results it suffices that $m > 1$, so that positive fees are feasible. Only when the platform sets its profit-maximizing fees do we take m sufficiently large that the optimum lies inside $\mathcal{S}$. The question of this section is how the platform's fee pair moves the adoption margin relative to laissez-faire.

Sellers' Rail Choice. Denote by

$$V_M(s_M, \xi) \equiv (1 - s_M) \pi_M\left(\frac{c}{1 - s_M}, \xi\right), \quad V_B(s_B) \equiv (1 - \omega)(1 - s_B) \pi_B\left(\frac{c}{(1 - \omega)(1 - s_B)}\right)$$

the profits a seller retains after paying the net fees on the two rails. Given the posted pair (s_M, s_B) , a seller of quality ξ accepts credit if and only if $V_B(s_B) \geq V_M(s_M, \xi)$, and money otherwise. Since V_M is strictly decreasing in ξ (Lemma 1) while V_B does not depend on ξ , this rule is a threshold rule: the threshold $\hat{\xi}(s_M, s_B)$ is defined by

$$V_B(s_B) = V_M(s_M, \hat{\xi})$$

whenever the crossing lies inside (ξ_ℓ, ξ_h) . Sellers above it take credit; sellers below it take money. Since the platform chooses the fees, corner solutions are possible: the threshold may fall outside the quality support, and one rail then empties. We focus on the interior case throughout, where both rails coexist.

Suppose $\hat{\xi}$ is interior. How does the threshold respond to the fees? Differentiating the indifference condition, $V'_B(s_B) = \partial_\xi V_M \cdot \partial \hat{\xi} / \partial s_B$; since $V'_B < 0$ and $\partial_\xi V_M < 0$ by Lemma 1, $\partial \hat{\xi} / \partial s_B > 0$. Similarly, $\partial_{s_M} V_M + \partial_\xi V_M \cdot \partial \hat{\xi} / \partial s_M = 0$ with $\partial_{s_M} V_M < 0$ gives $\partial \hat{\xi} / \partial s_M < 0$. Raising the credit fee pushes sellers back to money; raising the money fee pushes them into credit.

3.2 The Over-Supply Criterion

The platform over-supplies credit compared with the laissez-faire market exactly when $\hat{\xi} < \xi^*$. Fix the *reference seller*: the seller of quality ξ^* , who was exactly indifferent between the two rails under laissez-faire. She is the seller the fees can move between rails, so from here on we call her the **switcher**. Define the **log profit loss** that the platform's fee imposes on her in rail j as

$$C_M(s_M) \equiv \ln \frac{V_M(0, \xi^*)}{V_M(s_M, \xi^*)}, \quad C_B(s_B) \equiv \ln \frac{V_B(0)}{V_B(s_B)},$$

measured relative to the laissez-faire fees, which are zero on both rails. A fee pair is allocatively neutral exactly when the two losses are equal, $C_M = C_B$. Then clearly, at any fee pair for which the threshold is interior,

$$\hat{\xi} < \xi^* \iff C_M > C_B.$$

What goes into C_j ? Fix rail j and write $V_j(x)$ for the switcher's retained profit when the fee on that rail is x ; the loss C_j is the fall in $\ln V_j$ as x rises from the rail's laissez-faire anchor x_0 to its posted level x_1 . Two classical results track that fall. The envelope theorem gives $V_j'(x) = -R_j(x)$, where $R_j(x)$ is the switcher's sales revenue per unit of quality: a small fee increase costs her its base, her revenue. The Lerner condition $(z_j - a)/z_j = 1/\varepsilon_j$, with effective cost $a = c/(1 - x)$, converts revenue into profit: $R_j/V_j = \varepsilon_j/(1 - x)$, where ε_j is the switcher's demand elasticity in rail j at the price she posts under fee x . Combining the two,

$$-\frac{d \ln V_j}{dx} = \frac{\varepsilon_j}{1 - x} :$$

a small fee increase destroys a share of profit equal to the demand elasticity times the fall in the log retention rate $\ln(1 - x)$. An elastic seller earns a thin margin, so the same fee increase destroys more of her profit. To cumulate these losses, measure progress not by the fee x but by the *log-retention distance*

$$t = \ln \frac{1 - x_0}{1 - x},$$

which starts at $t = 0$ when $x = x_0$ and ends at $t = T_j \equiv \ln \frac{1 - x_0}{1 - x_1}$ when $x = x_1$. Since $dt = dx/(1 - x)$, the loss integrates to

$$C_j = \int_0^{T_j} \varepsilon_j(t) dt = \bar{\varepsilon}_j T_j,$$

where $\varepsilon_j(t)$ is the switcher's demand elasticity at the price induced by the fee at distance t , and $\bar{\varepsilon}_j \equiv \frac{1}{T_j} \int_0^{T_j} \varepsilon_j(t) dt$ is its average over the window. The loss is a rate times a distance: $\bar{\varepsilon}_j$ is the average erosion rate of profit, and T_j — call it the rail's **fee window** — is how far the fee drags

the seller.

The windows come out remarkably simple: the laissez-faire fee is zero on both rails, so the log-retention distance from $x_0 = 0$ to $x_1 = s_j$ is $T_j = \ln \frac{1}{1-s_j} = -\ln(1-s_j)$, $j = M, B$ — the net fee share in log form.

Theorem 1 *At any fee pair for which the threshold is interior, each log profit loss factors into an average elasticity and a fee window,*

$$C_j = \bar{\varepsilon}_j T_j, \quad T_j = \ln \frac{1}{1-s_j},$$

and consequently

$$\hat{\xi} < \xi^* \iff \frac{\bar{\varepsilon}_M}{\bar{\varepsilon}_B} > \frac{T_B}{T_M}.$$

The theorem is the most general statement of the section: it allows different fees on the two rails — hence different windows — and elasticities that differ across rails and move along each window as the fee rises. The direction is a race between two asymmetries: the elasticity ratio $\bar{\varepsilon}_M/\bar{\varepsilon}_B$, which demand determines, and the window ratio T_B/T_M , which the platform, or its regulator, posts. The rest of the section runs the race in two settings: equal windows first (Section 3.3), then the general fee pair/fee window (Section 3.4).

3.3 Equal Net Fee Setting

In the general model the platform may choose any pair (s_M, s_B) . We start from the equal net fee benchmark

$$s_M = s_B = s :$$

the platform charges the same net share on both rails, passing the cost of providing credit, ω , directly through to the users.

The benchmark does not describe every platform, but it arises in several real-world settings. Cost-based fee regulation permits fee differences across payment methods only to the extent that they reflect cost differences, so a compliant platform charges the same margin over each rail's cost. No-steering and parity rules similarly prevent the platform's margin from depending on how the buyer pays: cost recovery is allowed, but a rail-dependent markup is not. Many marketplaces already post a single ad valorem take rate by product category, independent of payment method, with payment costs recovered separately — exactly a common net share. And if the financing service is supplied competitively, the payment layer's own margin is competed away, leaving the platform with the same marketplace margin on both rails.

The equal net fee setting buys one simplification: the two fee windows are equal,

$$T_M = T_B = \ln \frac{1}{1-s} \equiv T,$$

so the direction hinges on the elasticities alone.

Proposition 3 *Under the equal net fee setting, for every share s at which the threshold is interior,*

$$C_M - C_B = \int_0^T [\varepsilon_M(t) - \varepsilon_B(t)] dt, \quad \text{hence} \quad \hat{\zeta}(s) < \zeta^* \iff \bar{\varepsilon}_M > \bar{\varepsilon}_B.$$

Two observations. First, the result holds at any equal fee level s — the platform's optimum, a regulated cap, any interior fee — because the level moves only the common window: the direction of the allocation is independent of the level of the fee. Second, this is where cost-based regulation is blind: equal net fees equalize everything a cost audit can inspect, but not what the fee does to the seller — a fee destroys a share of profit equal to the demand elasticity times the fee window, so with equal windows the more elastic money rail is taxed harder. A cost audit sees the costs; the direction of the allocation is decided by the elasticities, which no audit inspects.

The simplest case makes the point. Let F be Pareto, $\bar{F}(z) = (z/\theta_0)^{-\varepsilon}$ with $\varepsilon > 1$, and let H be arbitrary. Then $\lambda \equiv \varepsilon$ on both rails, so the switcher's credit elasticity is $\varepsilon_B(s) = \varepsilon$ at every s , while her money elasticity carries the extra term of Section 2.3, $\varepsilon_M(s) = \varepsilon + \gamma(p_M(s))$, where $\gamma(p) = p h(p) / \bar{H}(p)$ is the elasticity added by the liquidity constraint — the **liquidity elasticity** — evaluated at her money dollar price $p_M(s)$. Averaging over the common window,

$$\bar{\varepsilon}_M = \varepsilon + \bar{\gamma}_M > \varepsilon = \bar{\varepsilon}_B, \quad \bar{\gamma}_M \equiv \frac{1}{T} \int_0^T \gamma(p_M(t)) dt,$$

so by Proposition 3 the platform over-supplies credit — at every share s and for every liquidity distribution H .

With constant-elasticity demand, more is true: the entire allocation solves in closed form.

Proposition 4 *Let $\bar{F}(z) = (z/\theta_0)^{-\varepsilon}$ and let H be arbitrary. Under the equal net fee setting at any share s with $(1-s)\zeta^* > \zeta_\ell$,*

$$\hat{\zeta}(s) = (1-s)\zeta^*.$$

Proof. The threshold solves $V_B(s) = V_M(s, \hat{\zeta})$, and under constant elasticity both sides carry a common factor $(1-s)^\varepsilon$. On money, substituting $z = \tilde{z}/(1-s)$ into $V_M(s, \zeta) = \max_z [(1-s)z - c](z/\theta_0)^{-\varepsilon} \bar{H}(z\zeta)$ turns the maximization into the zero-fee problem of a seller of quality $\zeta/(1-s)$, scaled by $(1-s)^\varepsilon$, so $V_M(s, \hat{\zeta}) = (1-s)^\varepsilon V_M(0, \hat{\zeta}/(1-s))$. On credit, the net fee raises the effective

cost to $c/[(1-\omega)(1-s)]$, and $\pi_B(a) \propto a^{1-\varepsilon}$ gives $V_B(s) = (1-s)^\varepsilon V_B(0)$. The threshold condition therefore reduces to $V_M(0, \hat{\xi}/(1-s)) = V_B(0)$ — the laissez-faire indifference condition (2.2), whose unique solution is ξ^* (Lemma 1). Hence $\hat{\xi}/(1-s) = \xi^*$. ■

The threshold contracts by exactly the retention rate. The over-supplied band $[(1-s)\xi^*, \xi^*]$ depends only on G , the posted share, and the laissez-faire threshold; the liquidity distribution drops out entirely. Constant elasticity did the work of Proposition 4 by freezing λ : the two rails then read the same elasticity off the demand curve, and the elasticity gap of Proposition 3 was liquidity alone.

General F. Now consider a general F , under which the demand elasticity λ is no longer constant. At each point t of the common window the switcher's effective cost is $c e^t$ on money and $\frac{c}{1-\omega} e^t$ on credit; write $z_M(t)$ and $z_B(t)$ for the two prices those costs induce through the first-order conditions of Section 2.3, and let $p_M(t) \equiv z_M(t) \xi^*$ denote her money dollar price. Her elasticity in each rail is the generalized failure rate at her own price there, so the gap splits in two:

$$\bar{\varepsilon}_M - \bar{\varepsilon}_B = \underbrace{\frac{1}{T} \int_0^T \gamma(p_M(t)) dt}_{\bar{\gamma}_M, \text{ liquidity elasticity}} - \underbrace{\frac{1}{T} \int_0^T [\lambda(z_B(t)) - \lambda(z_M(t))] dt}_{\overline{\Delta\lambda}, \text{ the drift}}.$$

The first term is the liquidity elasticity: the extra elasticity the liquidity constraint adds on the money rail, and it is positive. The second term, the drift, is new: the two rails evaluate the same elasticity function λ at different prices, and the credit price is the higher one at every fee level — liquidity restrains the money price, and credit carries the higher effective cost (Section 2.3). Suppose Marshall's second law of demand holds: $\lambda(z)$ is nondecreasing in z , so demand is more elastic at higher prices. Then the credit seller sits at a more elastic point of the demand curve, the fee costs her more than a constant-elasticity benchmark would recognize, and the drift works against over-supply. If λ fell with the price, the drift would instead reinforce liquidity. The discussion above is summarized by the following proposition.

Proposition 5 *Let H be arbitrary. Under the equal net fee setting at any share s with interior threshold, $\hat{\xi}(s) < \xi^*$ if and only if $\bar{\gamma}_M > \max\{0, \overline{\Delta\lambda}\}$. The condition is automatic whenever λ decreases with the price, for then $\overline{\Delta\lambda} < 0$; with λ nondecreasing it reduces to $\bar{\gamma}_M > \overline{\Delta\lambda}$.*

If the credit cost is sufficiently small, over-supply becomes unconditional — the following corollary makes this precise.

Corollary 1 *Let λ be nondecreasing. The switcher is more elastic on money than on credit at every common effective cost: for every a ,*

$$\lambda(z_M(\xi^*, a)) + \gamma(p_M(a)) > \lambda(z^*(a)),$$

where $z_M(\xi^*, a)$ is the switcher's money price at effective cost a , $z^*(a)$ is the credit price at the same cost, and $p_M(a) \equiv z_M(\xi^*, a) \xi^*$ is her money dollar price. Consequently, at $\omega = 0$ over-supply holds for every F , every H , and every s , and by continuity for every sufficiently small ω .

Proof. Suppose the claim fails at some a . The Lerner condition writes each price as $a/z_j = 1 - 1/\varepsilon_j$, so the rail with the weakly lower elasticity charges the weakly higher price; here that means $z_M(\xi^*, a) \geq z^*(a)$. Since λ is nondecreasing, $\lambda(z_M) \geq \lambda(z^*(a)) \geq \lambda(z_M) + \gamma(p_M(a))$, contradicting $\gamma > 0$. At $\omega = 0$ the two rails face the same effective cost $c e^t$ at every $t \in [0, T]$, so the claim holds pointwise inside the integrals defining $\bar{\gamma}_M$ and $\bar{\Delta\lambda}$, giving $\bar{\gamma}_M > \bar{\Delta\lambda}$. ■

The corollary deserves a comment. It says that at any common effective cost, the switcher is more elastic on money than on credit: the liquidity constraint adds γ , and her price response hands part of the addition back but can never reverse it — a full reversal would mean she charges *more* than the credit seller at the same cost, while liquidity pushes her price the other way. The only force that can defeat liquidity is the credit cost ω itself, which moves the credit seller up the demand curve on its own account. At $\omega = 0$ that force is off, so over-supply holds for every F and H ; by continuity it survives while ω is small.

3.4 The General Pair (s_M, s_B)

Now let the platform post any pair (s_M, s_B) , so the two windows T_M and T_B may differ. To isolate the fee asymmetry, we keep the elasticity side at its sharpest: $\bar{F}$ with constant elasticity ε and H arbitrary. As in Section 3.3, $\varepsilon_B(t) \equiv \varepsilon$ and $\bar{\varepsilon}_M = \varepsilon + \bar{\gamma}_M$, with the switcher's average liquidity elasticity $\bar{\gamma}_M$ now taken over the money window, $\bar{\gamma}_M = \frac{1}{T_M} \int_0^{T_M} \gamma(p_M(t)) dt$. Theorem 1 reads

$$\hat{\xi} < \xi^* \iff n T_M > T_B, \quad n \equiv \frac{\varepsilon + \bar{\gamma}_M}{\varepsilon} = 1 + \frac{\bar{\gamma}_M}{\varepsilon} > 1.$$

An allocation-neutral schedule sets $C_M = C_B$, that is, $T_B = n T_M$. Substituting $T_j = -\ln(1 - s_j)$ gives a curve in the fee plane,

$$1 - s_B = (1 - s_M)^n;$$

$\bar{\gamma}_M$ varies with s_M through the window average, so the curve is defined implicitly, but nothing below needs more than $n > 1$.

Proposition 6 *Let $\bar{F}$ have constant elasticity. For any fee pair with interior threshold,*

$$\hat{\xi} < \xi^* \iff s_B < 1 - (1 - s_M)^n.$$

Proof. By Theorem 1, $\hat{\xi} < \xi^*$ iff $(\epsilon + \bar{\gamma}_M)T_M > \epsilon T_B$, i.e., $n T_M > T_B$, i.e., $-n \ln(1 - s_M) > -\ln(1 - s_B)$, which rearranges to $s_B < 1 - (1 - s_M)^n$. ■

Over-supply is therefore the entire set of fee pairs below the allocation-neutral curve (Figure 1). The equal-net-fee diagonal — the 45-degree line — lies strictly inside it. For small fees the curve is the ray $s_B \approx n s_M$, so neutrality requires the credit rail to bear a net share larger than the money rail's by exactly the switcher's elasticity ratio.

Platform's Profit-Maximizing Fees. Of particular interest is the pair that maximizes the platform's profit. The platform chooses $(s_M, s_B) \in \mathcal{S}$ to maximize

$$\Pi = s_M \bar{R}_M + (1 - \omega) s_B \bar{R}_B,$$

where $\bar{R}_M \equiv \int_{\xi_\ell}^{\hat{\xi}} R_M(s_M, \xi) \xi dG$ is money revenue and $\bar{R}_B \equiv \int_{\hat{\xi}}^{\xi_h} R_B(s_B) \xi dG$ is credit revenue; the factor $1 - \omega$ converts the credit net share into the platform's actual credit margin, because the resource cost ω is passed through.

We take m sufficiently large, so that the optimal fees are interior and characterized by the first-order conditions; with m close to one — a vanishing matching advantage — the participation constraint binds at the upper bounds of $\mathcal{S}$, the platform can charge only slightly positive fees, and we are back to the equal net fee setting.

The first-order conditions are

$$\bar{R}_M \left[1 - \frac{s_M k_M}{1 - s_M} \right] + B \frac{\partial \hat{\xi}}{\partial s_M} = 0, \quad (1 - \omega) \bar{R}_B \left[1 - \frac{s_B k_B}{1 - s_B} \right] + B \frac{\partial \hat{\xi}}{\partial s_B} = 0,$$

where $k_j \equiv \partial \ln \bar{R}_j / \partial \ln(1 - s_j) > 0$ is rail j 's tax-base elasticity and

$$B \equiv [s_M R_M(\hat{\xi}) - (1 - \omega) s_B R_B] \hat{\xi} g(\hat{\xi}).$$

Each fee thus works through two effects: a *revenue effect* — it changes what the platform collects from the sellers who stay on the rail — and a *boundary effect* — it moves the threshold $\hat{\xi}$, reassigning sellers between rails, with B measuring what the platform gains or loses when the switcher

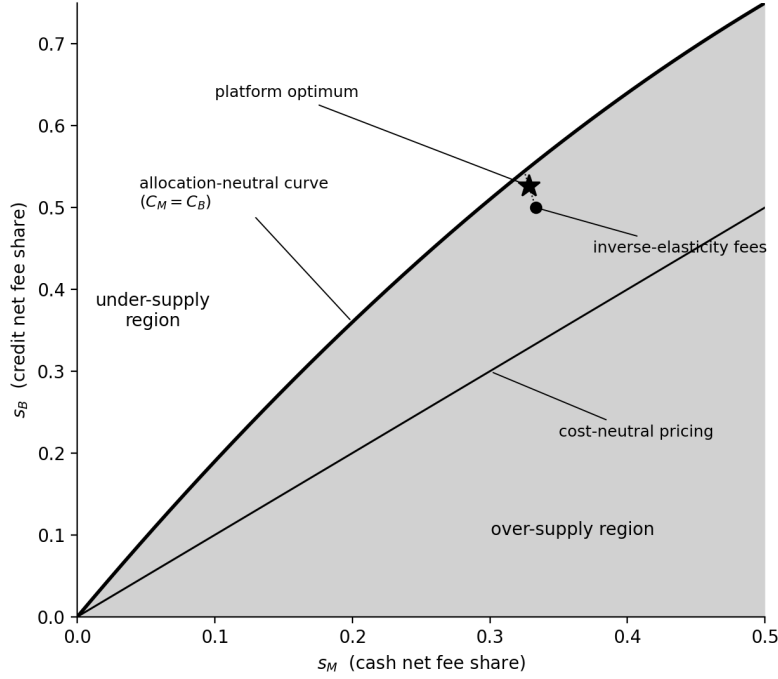


Figure 1: The fee plane at $\varepsilon = 2$.

Notes: The shaded area below the allocation-neutral curve $C_M = C_B$ is the over-supply region (Proposition 6); the area above it is the under-supply region. The 45-degree line — the equal net fee setting — lies inside the over-supply region at every fee level. The curve is drawn for $\bar{\gamma}_M = 2$ and a money-pool average $\bar{\gamma}_P = 1$, so $n = 2$: under exponential liquidity, where $\gamma(p)$ is proportional to the transaction size, this describes a switcher whose transaction is twice the pool's revenue-weighted average, and $\bar{\gamma}_P = 1$ is a normalization. The inverse-elasticity fees $(s_M^0, s_B^0) = (1/3, 1/2)$ price each rail off its own base (Section 3.4); the platform's optimum lies on the segment joining them to the curve (Proposition 7), here inside the over-supply region because $\bar{\gamma}_M = 2$ clears the bar in (3.1).

changes rails.

Suppose for a moment that the threshold did not move, $B = 0$. Then each fee would maximize the net revenue of its own rail, and the first-order conditions solve to

$$s_j^0 = \frac{1}{1 + k_j}, \quad j = M, B.$$

We call (s_M^0, s_B^0) **inverse-elasticity pricing**: the usual inverse-elasticity rule applied to the tax base $\bar{R}_j$, so the less elastic rail pays the higher fee. The credit rail is the less elastic base — its demand carries no liquidity elasticity — so inverse-elasticity pricing taxes credit harder, $s_B^0 > s_M^0$, and posts the larger credit window, $T_B > T_M$.

The profit-maximizing fees differ from inverse-elasticity pricing only because the boundary term is not zero, $B \neq 0$. The next proposition shows that the boundary term never reverses the

direction of the allocation — the sign under inverse-elasticity pricing is the sign at the optimum — so we can focus on inverse-elasticity pricing alone.

Proposition 7 *At an interior optimum with small fees, the sign of the misallocation is the same as under inverse-elasticity pricing. If inverse-elasticity pricing over-supplies credit, $C_M > C_B$, then $B > 0$, the optimal money fee is below its inverse-elasticity level, and the optimal credit fee is above it; the boundary term moves the fees toward the allocation-neutral curve but never across it. If inverse-elasticity pricing under-supplies credit, the corrections reverse.*

Proof. First write the boundary term in the units of the allocative comparison. At the switcher, $V_M = V_B \equiv V$, and the incidence identity from Section 3.2, $R_j/V_j = \varepsilon_j/(1 - x_j)$ — evaluated at $x_M = s_M$ and at $x_B = s_B$ — gives

$$s_M R_M(\hat{\xi}) = \frac{s_M \varepsilon_M}{1 - s_M} V, \quad (1 - \omega) s_B R_B = \frac{s_B \varepsilon_B}{1 - s_B} V.$$

Hence

$$B = V \hat{\xi} g(\hat{\xi}) [b_M - b_B], \quad b_j \equiv \frac{s_j \varepsilon_j}{1 - s_j}.$$

For small fees, $s_j/(1 - s_j) \approx T_j$ and the point elasticity ε_j is close to the window average $\bar{\varepsilon}_j$, so $b_j \approx \bar{\varepsilon}_j T_j = C_j$: the boundary term carries the sign of $C_M - C_B$.

Now the comparative statics: $\partial \hat{\xi} / \partial s_M < 0$ and $\partial \hat{\xi} / \partial s_B > 0$, as shown above. Suppose $C_M > C_B$, hence $B > 0$. Rearranging the first-order conditions,

$$1 - \frac{s_M k_M}{1 - s_M} = -\frac{B}{\bar{R}_M} \frac{\partial \hat{\xi}}{\partial s_M} > 0, \quad 1 - \frac{s_B k_B}{1 - s_B} = -\frac{B}{(1 - \omega) \bar{R}_B} \frac{\partial \hat{\xi}}{\partial s_B} < 0,$$

while inverse-elasticity pricing sets both left sides to zero. The optimum therefore lowers s_M and raises s_B relative to inverse-elasticity pricing, and both movements reduce $C_M - C_B$, which is increasing in s_M and decreasing in s_B . On the allocation-neutral curve $B = 0$, so the correction stops there; crossing the curve would reverse the sign of B and push the fees back. The case $C_M < C_B$ is symmetric. ■

The proposition is what makes inverse-elasticity pricing useful. We do not need to solve the full two-fee optimum; we only need to sign the misallocation under inverse-elasticity pricing, and the optimum lies on the same side of the allocation-neutral curve.

To obtain the criterion in closed form, let F be Pareto and H arbitrary. The credit side has $k_B = \varepsilon - 1$, so its inverse-elasticity window is

$$T_B = \ln \left(1 + \frac{1}{\varepsilon - 1} \right).$$

On the money side, $k_M = (1 - \psi)(\varepsilon - 1 + \bar{\gamma}_P)$, where $\psi = -d \ln z_M / d \ln \xi$ is the money seller's price restraint, an endogenous object pinned down by demand through the Lerner condition (Lemma 2), and $\bar{\gamma}_P$ is the revenue-weighted average of the liquidity elasticity $\gamma(p_M(\xi))$ across the sellers of the money pool, both treated as approximately constant across the pool; the inverse-elasticity money window is

$$T_M = \ln \left(1 + \frac{1}{k_M} \right).$$

The switcher's elasticities are $\bar{\varepsilon}_M = \varepsilon + \bar{\gamma}_M$ and $\bar{\varepsilon}_B = \varepsilon$, where $\bar{\gamma}_M$ is her own average liquidity elasticity over the money window, as defined at the start of this section. The two averages answer to different populations — $\bar{\gamma}_M$ to the one seller at the margin, $\bar{\gamma}_P$ to the whole money pool — and they coincide only when γ is constant. By Theorem 1 and Proposition 7, the platform's optimum over-supplies credit if and only if inverse-elasticity pricing does:

$$\bar{\gamma}_M > \varepsilon \left[\frac{\ln \frac{\varepsilon}{\varepsilon-1}}{\ln \left(1 + \frac{1}{(1-\psi)(\varepsilon-1+\bar{\gamma}_P)} \right)} - 1 \right]. \quad (3.1)$$

The condition now compares the two liquidity elasticities directly: the switcher's premium $\bar{\gamma}_M$ against a threshold built from the pool average $\bar{\gamma}_P$. A higher pool average raises the tax-base elasticity $k_M = (1 - \psi)(\varepsilon - 1 + \bar{\gamma}_P)$, shrinks the inverse-elasticity money window T_M , and pushes the threshold up. Over-supply therefore requires the switcher's liquidity elasticity to outweigh the fee differential that inverse-elasticity pricing posts off its pools.

Once $\gamma(p)$ rises with the transaction size p , two forces push the other way. The switcher posts the largest transactions in the money pool — she is its highest-quality seller — so $\bar{\gamma}_M > \bar{\gamma}_P$; and the price restraint $\psi > 0$ lowers k_M , widening the inverse-elasticity money window. The bar in (3.1) is then low. At $\varepsilon = 2$ and $\bar{\gamma}_P = 1$, over-supply requires $\bar{\gamma}_M > 1.42$ when $\psi = 0$, and only $\bar{\gamma}_M > 1.14$ with a modest price restraint of $\psi = 0.1$. Under exponential liquidity, $\bar{H}(p) = e^{-p/y_0}$, the example closes completely. The liquidity elasticity is $\gamma(p) = p/y_0$, linear in the transaction size, so $\bar{\gamma}_M = \bar{p}_M/y_0$ and $\bar{\gamma}_P = \bar{p}_P/y_0$: criterion (3.1) compares the switcher's average transaction with the pool's revenue-weighted average transaction. The price restraint is not a free parameter either — differentiating the money Lerner condition with respect to ξ pins it down as

$$\psi = \frac{\gamma}{\gamma + (\varepsilon + \gamma - 1)(\varepsilon + \gamma)},$$

evaluated at the switcher's transaction. At $\varepsilon = 2$, a switcher transaction of $\bar{\gamma}_M = 2$ against a pool average of $\bar{\gamma}_P = 1$ implies $\psi = 2/14 \approx 0.14$, and the bar in (3.1) falls to 1.02; the ratio $\bar{\gamma}_M/\bar{\gamma}_P = 2$ clears it, and a money pool spanning any realistic range of transaction sizes makes the switcher

a clear multiple of the pool average. In summary: the platform's own optimization preserves over-supply whenever the liquidity elasticity rises with the transaction size. Figure 1 illustrates the over-supply case: the platform's optimal fee pair falls inside the shaded region.

With a Pareto liquidity distribution, the comparison closes analytically. There $\gamma \equiv \alpha$ is constant and $\psi = 0$, so the switcher carries no premium: $\bar{\gamma}_M = \bar{\gamma}_P = \alpha$. Criterion (3.1) then fails for every parameter pair.

Corollary 2 *Let $\bar{F}$ and $\bar{H}$ both be Pareto, with $\lambda \equiv \varepsilon > 1$ and $\gamma \equiv \alpha > 0$. Under inverse-elasticity pricing — and hence, by Proposition 7, at the platform's optimum — $\hat{\xi} > \xi^*$: the platform under-supplies credit.*

Proof. With a Pareto liquidity distribution, $\gamma \equiv \alpha$ is constant, $\psi = 0$, and the switcher is representative of her pool, so $\bar{\gamma}_M = \bar{\gamma}_P = \alpha$. Criterion (3.1) specializes to

$$\alpha > \varepsilon \left[\frac{\ln \frac{\varepsilon}{\varepsilon-1}}{\ln \frac{\varepsilon+\alpha}{\varepsilon+\alpha-1}} - 1 \right],$$

which is equivalent to $C_M > C_B$, where $C_M = (\varepsilon + \alpha) \ln \frac{\varepsilon+\alpha}{\varepsilon+\alpha-1} = f(\varepsilon + \alpha)$ and $C_B = \varepsilon \ln \frac{\varepsilon}{\varepsilon-1} = f(\varepsilon)$, with $f(x) \equiv x \ln \frac{x}{x-1}$. Substituting $t \equiv 1/(x-1) > 0$ gives $f'(x) = \ln(1+t) - t < 0$, so f is strictly decreasing and $C_M = f(\varepsilon + \alpha) < f(\varepsilon) = C_B$. The criterion therefore fails, and by Proposition 7 the platform's optimum under-supplies credit: $\hat{\xi} > \xi^*$. ■

A Pareto liquidity distribution is therefore a special case, and a special one in a precise sense: constant γ makes the switcher exactly representative of her pool.

The message of this section extends beyond credit: a platform fee is a tax on elasticity, so any platform service that makes the demand its adopters face less elastic is over-supplied by the same logic. Amazon's delivery service illustrates the pattern: sellers who adopt it face less elastic demand, so by the same argument a platform supplying delivery at marginal cost over-supplies it relative to the free market.

Note also what the section did *not* need. Theorem 1 is silent about where a rail's elasticity comes from; under the benchmark restriction of Section 2 exactly one thing separates the two rails, the liquidity constraint that binds on money and not on credit. The tilt therefore rests on a single friction here — but the theorem will tilt the same way whenever anything else makes credit demand less elastic than money demand. Section 4 removes the restriction and supplies exactly such a second source: buyers who misperceive the cost of deferral make credit demand less elastic on their own account, and their naivety compounds the platform's tilt rather than substituting for it.

4 Consumer Naivety

The two preceding sections imposed $v = 1$: buyers perceived the load correctly, and the benchmark restriction let the visible benefit cancel it. We now drop both restrictions, so the load μ enters buyers' perceptions only partially and the benefit no longer cancels it. The restriction was never innocent: buyers report knowing that pay-later can be costly while being unable to state what it costs (Financial Conduct Authority, 2021; Consumer Financial Protection Bureau, 2022).

Naivety changes credit demand in two ways — its perceived level falls and its elasticity falls — and each change attaches to a different comparison: the first moves the market's adoption margin relative to the planner's, the second lands on the platform's fee. We rebuild pricing and adoption under naivety in Section 4.2, and take the naive market to the planner in Section 4.3 and to the platform in Section 4.4.

4.1 The Anchored Belief

A buyer who pays by credit faces the true all-in price $z^{\text{true}} = (1 - \beta + \mu)z$: she gives up the sticker price, gains the deferral benefit βz , and pays the load μz . She perceives it through (2.1), which we now write as

$$\tilde{z} = \phi z + \rho, \quad \phi \equiv 1 - \beta + v\mu, \quad \rho \equiv (1 - v)k. \quad (4.1)$$

We call ϕ the **marginal coefficient** — how much of a sticker-price dollar reaches perception — and ρ the **anchored part** — the component of the perceived price that no sticker-price change reaches. At $v = 1$ the anchored part vanishes and $\tilde{z} = z^{\text{true}}$, which with the benchmark restriction of Section 2 returns the demand of the two preceding sections.

Assumption 1 $v \in [0, 1]$ and $0 < k < \mu c / (1 - \omega)$.

The restriction on k says the stored impression falls short of the load at every equilibrium credit price, since $k < \mu c / (1 - \omega) < \mu z$ there. It joins the standing restrictions $\beta \in (0, 1)$ and $\mu \in (0, \beta)$ of Section 2; what we drop here is $v = 1$ and the benchmark restriction $\beta - \mu \rightarrow 0^+$, both of which now go.

Informally, a naive buyer misprices deferral in two steps. She sees only part of what deferral costs her, so credit looks cheaper than it is; and she fills the missing part not from the price in front of her but from memory — a stored impression of what pay-later costs on a purchase like

this one, which does not move when today's price moves. The first step is a mistake about the level of the price; the second is a mistake about its responsiveness.

The omission effect (a level mistake). The perceived price misses the true load by

$$z^{\text{true}} - \tilde{z} = (1 - v)(\mu z - k) > 0.$$

Buyers underestimate what credit costs, so they demand too much of it and the market's adoption margin understates the load.

The substitution effect (a slope mistake). The belief k does not move with the price, so a one-percent rise in z moves $\tilde{z}$ by less than one percent:

$$\frac{d \ln \tilde{z}}{d \ln z} = \frac{\phi z}{\phi z + \rho} < 1.$$

A more naive buyer has a lower ϕ and a larger ρ , hence a lower pass-through and strictly less elastic credit demand.

Away from the benchmark restriction the tie-break of Section 2 still holds, and for a stronger reason. Assumption 1 with $\mu < \beta$ gives $\tilde{z} < z$, since

$$\tilde{z} - z = -\beta z + v\mu z + (1 - v)k < -\beta z + v\mu z + (1 - v)\mu z = (\mu - \beta)z < 0.$$

Every buyer at a credit-accepting seller therefore prefers to defer: the visible benefit outweighs whatever load she perceives, whether or not she perceives all of it.

The opposite corner is worth naming now, because it carries the market result. At $v = 0$ the buyer relies entirely on the stored impression, $\tilde{z} = (1 - \beta)z + k$, and the load μ enters no demand curve, no price, and no profit — while the planner still subtracts it from every trade.

Why is k price-insensitive? A naive buyer does not compute what deferral will cost her on this purchase; she recalls what pay-later has cost her on purchases like it in the past. That memory is anchored to past prices: a seller's price change moves the price on the screen but not the memory — attribute substitution and anchoring in the sense of Tversky and Kahneman (1974), formally the linear case of sparse perception (Gabaix, 2014). The costs themselves invite such an encoding: a late fee is a flat amount and a missed-payment mark a discrete event, so the load hardly scales with the basket. And the memory is sticky, not permanently wrong: it forms from past prices and updates slowly, while the seller's price and the platform's fee move fast. The survey evidence

cited at the opening of the section fits this picture — cost beliefs that barely move with transaction size are exactly a sticky k .

4.2 Credit Pricing and Adoption under Naivety

Consider a seller who accepts credit at net fee s_B , so her effective cost is $a_B \equiv c/[(1-\omega)(1-s_B)]$. Buyers evaluate her price through (4.1), so her problem is

$$V_B(s_B; v) = (1-\omega)(1-s_B) \max_z (z - a_B) \bar{F}(\phi z + \rho).$$

The problem looks new but is not. Change variables to the perceived price. With $\tilde{z} = \phi z + \rho$ we have $z - a_B = [\tilde{z} - (\phi a_B + \rho)]/\phi$, so

$$V_B(s_B; v) = \frac{(1-\omega)(1-s_B)}{\phi} \max_{\tilde{z}} (\tilde{z} - \hat{a}) \bar{F}(\tilde{z}), \quad \hat{a} \equiv \phi a_B + \rho.$$

This is the standard monopoly problem — naivety does not change the shape of the problem. It relabels the seller's cost in perception units: the visible fraction of the load scales true cost by ϕ , and the anchored belief ρ is loaded on top, dollar for dollar, like a cost the seller must price over even though nobody pays it.

The point elasticity of credit demand with respect to the sticker price follows from the chain rule:

$$\varepsilon_B(z; v) = \underbrace{\lambda(\tilde{z})}_{\text{elasticity at the perceived price}} \times \underbrace{\frac{\phi z}{\phi z + \rho}}_{\text{pass-through}} = \lambda(\tilde{z}) \left(1 - \frac{\rho}{\tilde{z}}\right). \quad (4.2)$$

The second factor is below one because the anchored part does not move with z . A one-percent sticker-price increase raises the perceived price by less than one percent, so demand falls by less than the underlying elasticity would suggest. More naivety means lower pass-through and less elastic demand.

Writing $R_B(z) \equiv z \bar{F}(\tilde{z})$ for sticker revenue per unit of quality, pre-fee profit at the seller's optimum is $\pi_B(a_B; v) = (z_B - a_B) \bar{F}(\tilde{z}_B) = R_B(z_B)/\varepsilon_B(z_B; v)$: the rational formula, with naivety entering only through the elasticity in the denominator.

Credit profit is flat in quality and money profit strictly falls in it (Lemma 1), so the two cross at most once, exactly as in Section 2.3.

Proposition 8 Fix $v \in [0, 1]$ and suppose $V_M(0, \xi_h) < V_B(0; v) < V_M(0, \xi_\ell)$. Then there is a unique

$\xi^*(v) \in (\xi_\ell, \xi_h)$ solving

$$\underbrace{V_B(0; v)}_{\text{credit profit, flat in } \xi} = \underbrace{V_M(0, \xi^*(v))}_{\text{money profit, falling in } \xi}, \quad (4.3)$$

and every seller with $\xi > \xi^*(v)$ accepts credit while every seller with $\xi < \xi^*(v)$ takes money only.

How the threshold moves with what buyers see is the comparative static the rest of the paper uses.

Proposition 9 *Under Assumption 1:*

- (1) $V_B(0; v)$ is strictly decreasing in v , so $\xi^*(v)$ is strictly increasing wherever it is interior: more naivety means more credit adoption.
- (2) At $v = 0$, $V_B(0; 0)$ — and hence $\xi^*(0)$ — does not depend on μ .

Proof. (1) Apply the envelope theorem to $V_B(0; v) = (1 - \omega) \max_z (z - a_B) \bar{F}(\phi z + \rho)$ at $s_B = 0$, where $a_B = c/(1 - \omega)$. Only the direct effect of v on the perceived price survives:

$$\frac{\partial V_B}{\partial v} = (1 - \omega)(z_B - a_B) \bar{F}'(\bar{z}_B) \cdot \frac{\partial \bar{z}}{\partial v} \Big|_{z_B}, \quad \frac{\partial \bar{z}}{\partial v} \Big|_{z_B} = \mu z_B - k > 0,$$

where the inequality holds because $z_B > a_B$ and $k < \mu a_B$ by Assumption 1. Since $\bar{F}' < 0$, credit profit strictly falls as visibility rises. Money profit does not involve v , so in (4.3) the flat line shifts down while the declining curve stays put, and the crossing moves right.

(2) At $v = 0$, $\phi = 1 - \beta$ and $\rho = k$, so $\bar{z} = (1 - \beta)z + k$. The load enters no demand curve, no price, and no profit, and $\xi^*(0)$ therefore does not move with μ . ■

The intuition for part (1) is direct. Each step toward rationality raises the perceived price, cuts credit demand, and lowers credit profit, so fewer sellers find credit worthwhile. Part (2) is more subtle. At full naivety the hidden load is economically invisible to the entire market side: buyers do not price it, sellers' demand does not show it, and the adoption margin cannot respond to it however large μ is within $(0, \beta)$. Buyers pay it all the same. The one ledger that still carries μ is the planner's, and that is the comparison we turn to next.

4.3 The Naive Market against the Planner

The planner of Section 2.4 chooses which sellers accept credit and can do nothing else. With naive buyers the welfare calculation changes on the credit side alone.

A seller assigned to money prices at z_M , and a buyer trades only if she wants the good and can pay for it, so surplus per unit of quality is $W_M(\xi) = \bar{H}(p_M) \int_{z_M}^{\theta_h} (\theta - c) dF(\theta)$, exactly as before.

A seller assigned to credit prices at $z_B(v)$, and here the calculation needs two prices. *Selection runs on the perceived price*: a buyer trades if $\theta \geq \tilde{z}_B(v)$. *Valuation runs on the true price*: each trade delivers the buyer $\theta - (1 - \beta + \mu)z_B$ while the seller keeps $(1 - \omega)z_B - c$, so each trade creates $\theta - c - \kappa z_B$ with $\kappa \equiv \omega + \mu - \beta$. Therefore

$$W_B(v) = \int_{\tilde{z}_B(v)}^{\theta_h} [\theta - c - \kappa z_B(v)] dF(\theta), \quad \kappa \equiv \omega + \mu - \beta. \quad (4.4)$$

The composite κ is the net social cost of a credit dollar: the rail consumes ω , the load destroys μ , and deferral creates β . Since $\mu < \beta$, this cost is negative whenever $\omega < \beta - \mu$, in which case a credit dollar is socially cheap and the planner leans toward credit on its own account.

The two prices in (4.4) divide credit buyers into two groups. Buyers with $\theta \geq (1 - \beta + \mu)z_B$ have positive true surplus and should buy; their trades add to welfare. Buyers with $\tilde{z}_B \leq \theta < (1 - \beta + \mu)z_B$ have positive perceived surplus and negative true surplus; they buy and regret it, and their trades subtract from welfare. The regret interval is nonempty whenever $v < 1$, with width

$$(1 - \beta + \mu)z_B - \tilde{z}_B = (1 - v)(\mu z_B - k) > 0,$$

and it closes at $v = 1$. This interval is the welfare content of naivety.

Why does it create market over-provision? Because the seller and the buyer act on the same wrong price and the planner does not. The buyer chooses credit because her perceived price is too low; the seller accepts credit because demand looks too high. The regret buyers bridge the two: they make credit look profitable to the seller and costly to the planner, so the market's adoption margin is pushed toward credit by exactly the buyers the planner would rather not see there.

The planner's threshold $\xi^{**}(v)$ solves $W_M(\xi^{**}) = W_B(v)$. Existence and uniqueness need one condition on the money side.

Lemma 3 $W_B(v)$ is flat in ξ . Let $\psi \equiv -d \ln z_M / d \ln \xi$ be the rate at which the money seller cuts her normalized price as quality rises. If

$$\psi < \frac{\gamma(p_M)}{\gamma(p_M) + \lambda(z_M)},$$

then W_M is strictly decreasing in ξ and the planner's threshold $\xi^{**}(v)$ is unique whenever interior.

Proof. Flatness of W_B : with v constant, neither z_B nor $\tilde{z}_B$ involves ξ , so (4.4) does not either.

For W_M , since $d \ln p_M / d \ln \xi = 1 - \psi$, differentiating $W_M = \bar{H}(p_M) \int_{z_M}^{\theta_h} (\theta - c) dF(\theta)$ gives

$$\frac{d \ln W_M}{d \ln \xi} = -\gamma(p_M)(1 - \psi) + \frac{z_M(z_M - c)f(z_M)}{\int_{z_M}^{\theta_h} (\theta - c) dF(\theta)} \psi.$$

The first term is the screening effect, negative whenever $\psi < 1$. The second is the price-restraint effect, with the sign of ψ and coefficient at most $\lambda(z_M)$ because $\int_{z_M}^{\theta_h} (\theta - c) dF(\theta) \geq (z_M - c)\bar{F}(z_M)$. For $\psi \leq 0$ both terms are nonpositive and the first is strictly negative. For $\psi > 0$ the sum is at most $-\gamma(1 - \psi) + \lambda\psi$, which is negative exactly under the stated condition. ■

Two forces act on money surplus as quality rises. A higher ξ raises the dollar price and shuts out more cash-poor buyers — the liquidity effect, which lowers money surplus. A higher ξ may also make the seller cut her normalized price, which benefits the buyers who remain — the price-restraint effect, which raises it. Lemma 3 asks that liquidity effect dominate.

The surplus race. Write $S(z) \equiv \int_z^{\theta_h} (\theta - z) dF(\theta)$ for buyer surplus at price z . On the credit rail selection uses the perceived price while true buyer surplus uses the true all-in price, so

$$S_B^{\text{true}}(v) \equiv \int_{z_B(v)}^{\theta_h} [\theta - (1 - \beta + \mu)z_B(v)] dF(\theta).$$

At the laissez-faire threshold profits are equal across rails. Splitting each surplus into buyer surplus plus seller profit, the profit terms cancel and $W_M(\xi^*(v)) - W_B(v) = \bar{H}(p_M^*)S(z_M) - S_B^{\text{true}}(v)$, where $p_M^* \equiv z_M \xi^*(v)$. By Lemma 3 this difference crosses zero once, from above, at $\xi^{**}(v)$, so

$$\xi^*(v) < \xi^{**}(v) \iff \bar{H}(p_M^*)S(z_M) > S_B^{\text{true}}(v). \quad (4.5)$$

This is the race of (2.3) with one change: credit buyer surplus is selected at the perceived price and valued at the true one. The economics is unchanged. Credit expands the market, serving every buyer who wants the good while money reaches only the fraction $\bar{H}(p_M^*) < 1$ who can pay on the spot; credit also carries a higher true price, so each credit sale delivers less true buyer surplus. Laissez-faire over-provides credit exactly when the expansion outweighs the price increase.

Naivety works on both margins of the credit side. It lowers the perceived price, which brings more buyers onto the credit rail, so the extensive margin expands. It does not lower the true price those buyers pay — under the anchored belief it can raise it — so the intensive margin does not improve. The market sees the larger demand and finds credit more profitable; the planner sees the same buyers and subtracts the true load from every trade.

Proposition 10 *Maintain Assumption 1 and the condition of Lemma 3, and let the thresholds be interior.*

- (1) *At $v = 0$ the market threshold $\xi^*(0)$ does not depend on μ , while $W_B(0)$ is strictly decreasing in μ and hence $\xi^{**}(0)$ is strictly increasing in μ . The gap $W_M(\xi^*(0)) - W_B(0)$ is therefore strictly increasing in μ , and if μ is large enough within $(0, \beta)$ the fully naive market over-provides credit: $\xi^*(0) < \xi^{**}(0)$.*
- (2) *Fix such a μ . Then there exists $v^\dagger > 0$ such that $v < v^\dagger$ implies $\xi^*(v) < \xi^{**}(v)$.*

Proof. (1) The first claim is Proposition 9(2). For the second, fix $v = 0$: the perceived price is $(1 - \beta)z + k$, so $z_B(0)$ and $\tilde{z}_B(0)$ do not involve μ , and μ enters (4.4) only through κ , giving

$$\frac{\partial W_B(0)}{\partial \mu} = -z_B(0) \bar{F}(\tilde{z}_B(0)) < 0.$$

W_M does not involve μ . Under Lemma 3, $W_M - W_B$ crosses zero once from above at $\xi^{**}(0)$, so lowering the flat line W_B shifts the crossing right and $\xi^{**}(0)$ strictly increases in μ . Since $\xi^*(0)$ is fixed while $W_M(\xi^*(0)) - W_B(0)$ rises continuously and strictly in μ , the sign turns positive once μ is large enough, provided that point lies within $(0, \beta)$.

(2) All objects are continuous in v : $z_B(v)$ and $\tilde{z}_B(v)$ by the maximum theorem, $V_B(0; v)$ with them, hence $\xi^*(v)$ through (4.3) and $\xi^{**}(v)$ through $W_M(\xi^{**}) = W_B(v)$, using Lemma 1 and Lemma 3 for strict monotonicity in ξ . The gap $\xi^{**}(v) - \xi^*(v)$ is therefore continuous in v and strictly positive at $v = 0$, so it remains strictly positive on $[0, v^\dagger)$ for some $v^\dagger > 0$. ■

Part (1) says the load decides whether the gap exists. At full naivety the market ignores μ while the planner subtracts it from every credit trade, so the market's margin sits still as the planner's rises. A small μ may not move the planner enough to matter; a large enough μ makes the planner refuse sellers the market still finds profitable.

Part (2) says visibility decides how large the gap is. As v rises the perceived price moves toward the true price, the regret interval $(1 - v)(\mu z_B - k)$ shrinks, and the market's demand curve starts to carry the cost information the planner uses. At $v = 1$ the two prices coincide, there are no regret buyers, and the informational gap closes; whatever difference remains between the market and the planner is the liquidity comparison of Section 2.4, not naivety.

Proposition 10(2) is therefore the education result. A cutoff $v^\dagger$ is a visibility level at which the two thresholds cross, $\xi^*(v^\dagger) = \xi^{**}(v^\dagger)$. Below it the market over-provides credit; above it the informational gap closes. Continuity guarantees at least one such crossing once over-provision holds at $v = 0$, and uniqueness follows when the gap is monotone in v .

4.4 Naivety and the Platform's Tilt

We now return to the comparison of Section 3, between the platform's threshold $\hat{\zeta}$ and the market's ζ^* . Everything in that section survives: Theorem 1 factors each rail's log profit loss into $C_j = \bar{\varepsilon}_j T_j$, and the direction of the tilt is the race between the elasticity ratio and the window ratio. Naivety enters at one point only — it lowers $\bar{\varepsilon}_B$.

General demand. The central comparative static needs no distributional assumption. It needs only that the anchored belief be price-insensitive.

For arbitrary F the credit seller solves the standard monopoly problem at perceived effective cost $\hat{a} = \phi a_B + \rho$. Let $w^*(\cdot)$ denote the standard monopoly price, so $\tilde{z}_B = w^*(\hat{a})$ and the point elasticity is (4.2). Two monotonicity assumptions make the comparative static exact: the standard monopoly price is nondecreasing in cost, $w^{*'}(\cdot) \geq 0$, and the generalized failure rate is nondecreasing, $\lambda'(\cdot) \geq 0$. Both hold under Pareto, and the second is Marshall's second law of demand.

Proposition 11 *Let $w^{*'}(\cdot) \geq 0$ and $\lambda'(\cdot) \geq 0$. Under Assumption 1, at every point t of the credit window, $\partial \varepsilon_B(t; v) / \partial v > 0$. Consequently $\bar{\varepsilon}'_B(v) > 0$, and under the equal net fee setting*

$$\frac{d}{dv}(C_M - C_B) = -T \bar{\varepsilon}'_B(v) < 0.$$

If the platform over-supplies credit at visibility v_1 , it over-supplies credit at every $v < v_1$, and the wedge $C_M - C_B$ is strictly larger.

Proof. Differentiate $\hat{a} = \phi a_B + \rho$. Since $\partial \phi / \partial v = \mu$ and $\partial \rho / \partial v = -k$,

$$\frac{d\hat{a}}{dv} = \mu a_B - k > 0$$

by Assumption 1. With $w^{*'} \geq 0$ the perceived price $\tilde{z}_B = w^*(\hat{a})$ is nondecreasing in v , and with $\lambda' \geq 0$ the first factor of (4.2) is nondecreasing in v . The second factor is $1 - \rho / \tilde{z}_B$, whose derivative is positive because ρ falls with v while $\tilde{z}_B$ does not fall. Hence the product strictly increases in v at every t , and integrating over the window gives $\bar{\varepsilon}'_B(v) > 0$. Under the equal net fee setting the two windows are equal, $T_M = T_B = T$, so $C_M - C_B = T[\bar{\varepsilon}_M - \bar{\varepsilon}_B(v)]$ and $\bar{\varepsilon}_M$ does not involve v . ■

Naivety lowers the credit elasticity for a single reason: the anchored belief withholds part of every sticker-price change from perception. Pareto demand is needed only to make the two elasticities explicit and to deliver exact thresholds, which is what we do next.

Proposition 12 *Let $\bar{F}$ be Pareto with elasticity ε and let H be arbitrary. Under the equal net fee setting at any share $s > 0$ with interior thresholds:*

- (1) $C_M - C_B = T[\bar{\varepsilon}_M - \bar{\varepsilon}_B(v)] = T[\varepsilon + \bar{\gamma}_M - \bar{\varepsilon}_B(v)] > 0$, so $\hat{\xi}(s, v) < \xi^*(v)$ for every $v \in [0, 1]$.
- (2) $\frac{d}{dv}(C_M - C_B) = -T\bar{\varepsilon}'_B(v) < 0$: the wedge that pushes the switcher into credit is strictly larger the more naive consumers are.

Proof. (1) Under Pareto $\bar{F}$, $\lambda \equiv \varepsilon$, so $\bar{\varepsilon}_M = \varepsilon + \bar{\gamma}_M$ as in Section 3.3, and (4.2) gives $\varepsilon_B(t; v) = \varepsilon(1 - \rho/z_B) \leq \varepsilon$ at every t , hence $\bar{\varepsilon}_B(v) \leq \varepsilon$. Since $\bar{\gamma}_M > 0$, the bracket is strictly positive, and Theorem 1 converts $C_M > C_B$ into $\hat{\xi} < \xi^*$. (2) Immediate from Proposition 11, since $\bar{\varepsilon}_M$ does not depend on v . ■

The proposition separates cleanly into what the liquidity effect does and what naivety adds. The platform over-supplies credit *at every* v , including $v = 1$, because the liquidity constraint alone makes money the more elastic rail and a common margin therefore takes a larger share of money profit than of credit profit. Naivety does not create this tilt. What naivety does is mute the one force that limits it: the elasticity of credit demand. The anchored belief is the part of the perceived credit price that the fee — passed through the seller's sticker price — cannot reach; the larger that part, the less elastic credit demand, the more gently the fee treats the credit rail, and the further the switcher's comparison tilts.

Exact contraction under Pareto liquidity. With Pareto H the tilt becomes a formula. Since $\gamma \equiv \alpha$, the money price does not depend on quality and retained money profit factors as $V_M(x, \xi) = \Phi(x)\xi^{-\alpha}$ with $\Phi(x) \propto (1-x)^{\varepsilon+\alpha}$, so $C_M = (\varepsilon + \alpha)T$. The two indifference conditions — laissez-faire, $V_B(0; v) = \Phi(0)(\xi^*)^{-\alpha}$, and platform, $V_B(s; v) = \Phi(s)\hat{\xi}^{-\alpha}$ — divide into $\ln[\xi^*(v)/\hat{\xi}(s, v)] = (C_M - C_B)/\alpha$.

Proposition 13 *Let $\bar{F}$ and $\bar{H}$ both be Pareto. Under the equal net fee setting at share s with interior thresholds,*

$$\frac{\hat{\xi}(s, v)}{\xi^*(v)} = (1-s)^{1 + \frac{\varepsilon - \bar{\varepsilon}_B(v)}{\alpha}},$$

with $\bar{\varepsilon}_B(v)$ given by (4.2). The exponent is strictly decreasing in v ; at $v = 1$ it equals one, recovering the rational benchmark $\hat{\xi} = (1-s)\xi^$ of Proposition 4, and for every $v < 1$ the contraction is strictly deeper.*

The formula rewards a slow reading. The rational contraction $1 - s$ is the pure screen effect: a fifteen-percent take rate moves the adoption threshold down fifteen percent, whatever the liquidity distribution. Naivety adds the premium $[\varepsilon - \bar{\varepsilon}_B(v)]/\alpha$ to the exponent — the gap between

what the credit elasticity would be and what the anchored belief lets it be, scaled by the strength of the screen. Note what the statement is and is not about: both $\hat{\xi}$ and ξ^* move with v , and the claim concerns their *ratio*. Naivety deepens what the platform adds on top of whatever the naive market already does.

General willingness to pay. For arbitrary F the money elasticity remains $\lambda(z) + \gamma(p)$, Theorem 1 still governs the direction, and the integrand now splits into three parts:

$$\varepsilon_M - \varepsilon_B = \underbrace{\gamma(p_M)}_{\text{screen}} + \underbrace{\lambda(\tilde{z}_B) \frac{\rho}{\tilde{z}_B}}_{\text{naivety discount}} - \underbrace{[\lambda(\tilde{z}_B) - \lambda(z_M)]}_{\text{drift}}. \quad (4.6)$$

The screen is positive, as always. The naivety discount is the new term, and it is nonnegative, zero only at $v = 1$: whatever the shape of demand, the anchored belief withholds part of every sticker-price change from perception, and that always lowers the credit elasticity. The drift can push back. The two rails read the same $\lambda(\cdot)$ at different prices, and under Marshall's second law a credit rail sitting at a higher perceived price is a more elastic point of the curve. Compare this with the rational decomposition of Section 3.3, where the screen fought the drift alone. Naivety now fights on the screen's side.

Proposition 14 *Let λ and $w^*(\cdot)$ be nondecreasing, let H be arbitrary, and suppose $(1 - v)k \leq (\beta - v\mu)c$. Then at $\omega = 0$, for every such F , every H , every s with interior thresholds, and every v , the platform over-supplies credit; by continuity the conclusion survives for all sufficiently small $\omega > 0$.*

Proof. At $\omega = 0$ the two rails share the effective cost $a = c e^t$ at every point of the common window. The stated condition gives $\hat{a} = \phi a + \rho \leq \phi a + (\beta - v\mu)a = a$ at every $t \geq 0$, so the credit seller's perceived cost lies below the money seller's true cost, and $\tilde{z}_B = w^*(\hat{a}) \leq w^*(a)$ because w^* is nondecreasing. The equal-cost comparison of Corollary 1 gives $\lambda(z_M) + \gamma(p_M) > \lambda(w^*(a))$ at any common cost a . Chaining these, pointwise in t ,

$$\varepsilon_M(t) = \lambda(z_M) + \gamma(p_M) > \lambda(w^*(a)) \geq \lambda(\tilde{z}_B) \geq \lambda(\tilde{z}_B) \left(1 - \frac{\rho}{\tilde{z}_B}\right) = \varepsilon_B(t; v),$$

using λ nondecreasing for the middle step and the pass-through bound for the last. Integrating over the window gives $C_M > C_B$, and Theorem 1 concludes. Continuity in ω extends the strict inequality to small ω . ■

The side condition $(1 - v)k \leq (\beta - v\mu)c$ asks that the mistaken belief not be large relative to the visible deferral benefit; $k \leq (\beta - \mu)c$ is a simple sufficient version. It can be dropped at the cost of a longer argument bounding the drift directly. Outside the Pareto family, then, over-

supply holds under the same “credit is not too expensive” proviso as in Corollary 1, and naivety only makes that proviso easier to satisfy.

4.5 The Fee Plane with Naive Buyers

Return to Pareto $\bar{F}$ and let the platform post any pair (s_M, s_B) . By Theorem 1, over-supply occurs if and only if $(\varepsilon + \tilde{\gamma}_M)T_M > \bar{\varepsilon}_B(v)T_B$, that is,

$$s_B < 1 - (1 - s_M)^{n(v)}, \quad n(v) \equiv \frac{\varepsilon + \tilde{\gamma}_M}{\bar{\varepsilon}_B(v)} > 1. \quad (4.7)$$

The boundary is the **allocation-neutral curve** of Section 3.4, now indexed by visibility. For small fees it is the ray $s_B \approx n(v)s_M$: neutrality requires the credit rail to bear a net share *larger* than the money rail’s, by the switcher’s elasticity ratio. Since $\bar{\varepsilon}'_B(v) > 0$ we have $n'(v) < 0$, so naivety enlarges the over-supply region. A schedule engineered to be neutral against rational buyers still over-supplies credit against naive ones, and the neutral credit surcharge a regulator would need grows with naivety — a behavioral quantity that no cost audit, and no demand estimation on money transactions, can reveal.

The Pareto–Pareto optimum. Corollary 2 showed that under Pareto–Pareto *and* inverse-elasticity pricing the rational platform under-supplies credit, because constant γ makes the switcher exactly representative of her pool. Naivety pushes against that conclusion from both sides of (4.7): it lowers $\bar{\varepsilon}_B(v)$, which raises $n(v)$ and enlarges the over-supply region, and it also lowers the credit tax-base elasticity k_B , which moves the inverse-elasticity fees themselves. Signing the net effect requires recomputing k_B under naivety, which we do not do here. The equal-net-fee results of Section 4.4 are unaffected: they hold at every share whenever the platform is constrained to a common net margin.

5 Conclusion

We modelled buy-now-pay-later as a payment rail with two defining features. It relieves a liquidity constraint, so a buyer without the cash on hand can still trade. And it carries a hidden load — late fees, penalty charges, credit-record damage — that the buyer pays in full and perceives in part.

The first feature alone settles the platform question. A money sale must clear a liquidity hurdle that a credit sale skips, so money demand is the more elastic rail. An *ad valorem* fee is

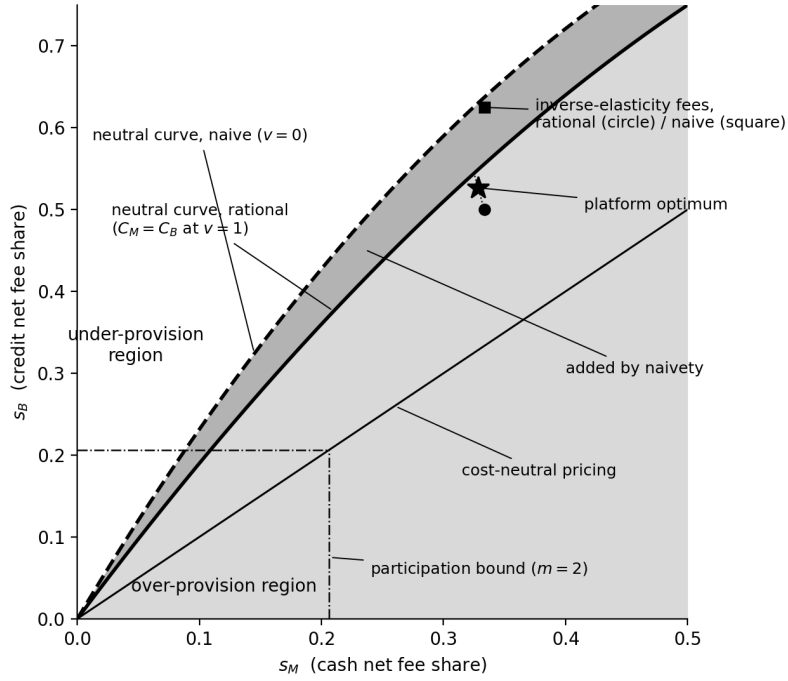


Figure 2: The fee plane with naive buyers.

Notes: The plane is drawn at $\varepsilon = 2$, switcher liquidity elasticity $\bar{\gamma}_M = 2$, and money-pool average $\bar{\gamma}_P = 1$. The solid neutral curve is drawn for rational buyers, $\bar{\varepsilon}_B(1) = 2$, so $n(1) = 2$; the dashed curve for naive buyers with $\bar{\varepsilon}_B(0) = 1.6$, so $n(0) = 2.5$. Everything below a neutral curve over-supplies credit at that visibility level, and the dark band is the set of schedules neutral for rational buyers but distortionary for naive ones. The equal-net-fee diagonal lies inside the over-supply region at every fee level and every v . The circle and square mark the inverse-elasticity fees for $v = 1$ and $v = 0$, and the star marks the platform optimum, on the segment from the inverse-elasticity pair toward the neutral curve (Proposition 7). The dash-dotted box is the participation bound $s \leq 1 - m^{-1/(\varepsilon+\alpha)}$ at $m = 2$ and $\varepsilon + \alpha = 3$: the fees an actual platform can post are capped by its matching advantage.

levied on revenue and paid out of profit, so the share of profit it destroys is the seller's demand elasticity. A platform charging the same net margin on both rails therefore taxes money harder than credit and tilts sellers onto the credit rail — at every fee level, for every liquidity distribution, and with fully rational buyers. Cost-based regulation is blind to this: equal net fees equalize everything a cost audit can inspect, and the direction of the allocation is decided by elasticities, which no audit inspects.

The second feature adds a second distortion and amplifies the first. Buyers omit part of the load and substitute a stored, price-insensitive impression in its place, and the two mistakes do different work. The omission is a level mistake: it removes the load from every market ledger while leaving it on the planner's, so a heavy enough load pushes the market's adoption mar-

gin past the planner's. The substitution is a slope mistake: the stored impression is the part of the perceived credit price that a fee, passed through the sticker price, can never reach, so naive credit demand is inelastic to exactly the instrument the platform wields. The screen creates the platform's tilt; naivety subsidizes it.

Two distortions with two causes call for two instruments. Consumer education raises v and closes the informational margin, but only disclosure that is per-transaction and price-denominated does so — a generic warning moves the stored impression, not its responsiveness to the price on the screen. Education cannot close the platform margin: the tilt survives at $v = 1$, where the wedge is the pure liquidity term. That margin is priced in the platform's own coin and closes in that coin, through the structure of net shares across the rails rather than a cap on their level. The subsidy corner matters most here: platforms routinely price credit below cost to drive merchant adoption, and a rule capping only positive credit fees never reaches it. The regulatory wave arriving in the United Kingdom, the European Union, and Australia is a well-aimed attack on the informational margin. The other one is still unattended.

The mechanism is not specific to credit. A platform fee is a tax on elasticity, so any platform service that makes the demand its adopters face less elastic is over-supplied by the same logic — even by a platform that supplies the service at cost.

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